

GALA: Anatomy-Aware Language–Motion Alignment for Efficient Text-to-Motion Rectified Flow

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HumanML3D test, $n=4544$, 20 replications · Guo / MDM protocol

Code: <https://github.com/yanghxx/gala-motion>

Abstract

Text-to-motion generation must couple language semantics with skeletal kinematics, yet most latent generators either pool a single CLIP vector against a single motion code or denoise raw poses at a high function-evaluation cost. Existing latent T2M methods lack an explicit structural bridge between skeletal topology, fine-grained language semantics, and the generative trajectory. GALA establishes that bridge: a skeleton-graph VAE exposes five body-part latents; learnable anatomical queries align CLIP tokens to those parts; a rectified-flow DiT then transports noise under motion-space kinematic constraints, sampled with a 20–50 step Euler ODE. On official HumanML3D, the current public checkpoint—graph tokenizer, global InfoNCE, no part queries and no kinematic loss—reaches Top-3 R-Precision 0.768 at 20 NFEs and FID 0.312 at 50 NFEs, beating MDM on every reported metric on one RTX 4060 (8 GB). We isolate part alignment and kinematic decoding as separate ablations rather than attributing these headline numbers to the full system. The remaining gap to masked-token SOTA is isolated to the generator, not the tokenizer (reconstruction FID 0.003).

1 Introduction

Text-to-motion (T2M) maps a sentence to a variable-length pose clip [1, 2]. HumanML3D [1] and KIT-ML [3] are the standard splits; the Guo protocol—R-Precision, FID [4], multimodal distance, Diversity, 20 replications—is the number reviewers compare [1, 2]. Diffusion on raw 263-D poses [2, 5] is faithful but slow; latent diffusion [6] and discrete tokens [7–9] push FID down at the cost of a codebook or a long denoising chain. Rectified flow [10, 11] replaces that chain with a straight ODE, which is the right inductive bias when the budget is a single 8 GB GPU.

A topology-aware variational tokenizer already yields a compact, near-lossless motion latent, so generation reduces to learning a *straight* path in that space that is explicitly aligned with language. Existing latent T2M methods lack an explicit structural bridge between skeletal topology, fine-grained language semantics, and the generative trajectory. GALA establishes this bridge by linking three stages that are not interchangeable: the graph tokenizer exists to expose anatomical parts; part-level alignment exists so language can address those parts; kinematic flow exists because reconstruction FID shows that fidelity is lost in the ODE, not in the VAE—skeleton graph $\rightarrow$ part latents

$\leftrightarrow$ language tokens $\rightarrow$ kinematic rectified flow.

Contributions.

1. **Anatomy-aware motion tokenizer.** A CTR graph encoder plus stride-4 VAE maps 263-D poses to 256-D latents at 5 Hz, preserving skeletal dependencies and exposing part-structured features. Val reconstruction FID is 0.003.
2. **Fine-grained language–motion alignment.** Instead of pooled InfoNCE alone, GALA forms token-to-body-part correspondences so that a phrase such as “raises the left arm” conditions the matching limb latent.
3. **Kinematic-aware rectified flow.** An 8-block DiT learns a straight latent ODE under classifier-free guidance [12], with velocity, bone-length, and foot-contact penalties in decoded motion space, enabling 20–50 NFE sampling.

We compare against native-HumanML3D generators (MDM, MLD, M2DM, T2M-GPT, EMDM, MoMask) and treat GENMO [13] as one generalist SMPL baseline, not as the target.

2 Related work

Pose-space DDPMs [2, 5, 14] are faithful but slow. MLD [6] diffuses in a VAE latent—the same split we use, without a skeleton graph or a straight flow. T2M-GPT [7], M2DM [9], MoMask [8], and MMM [15] quantize motion; we stay continuous because a 256-D VAE already reconstructs at FID 0.003. Consistency models [16] and EMDM [17] cut the denoising chain; rectified flow [10, 11] learns a straight path that a DiT [18] can parameterize under CFG [12]. ST-GCN [19] and CTR-GCN [20] supply the topology prior. Part-level T2M [21, 22] generates or attends over limbs; GALA instead aligns language tokens to part latents inside one continuous flow. GENMO [13] is a generalist SMPL [23] model whose HumanML3D numbers pass through a conversion the authors call an inherent disadvantage; we list it as a reference, not as the target.

3 Method

Representation. A clip is $x_{1:T} \in \mathbb{R}^{T \times 263}$, $T \leq 196$, 20 fps, Guo features [1]: root motion, local joints, velocities, and four foot-contact channels. KIT-ML uses the analogous 251-D,

21-joint feature. Text is frozen CLIP ViT-B/32 [24] tokens $c_{1:N} \in \mathbb{R}^{N \times 512}$.

Part-aware graph tokenizer. Local joint positions and velocities form a J -node graph. Two CTR blocks mix a static anatomical adjacency with a sample-dependent attention graph; an ST-GCN variant (static adjacency only) and a graph-free conv VAE are the tokenizer controls. Joints are pooled into $P=5$ parts $\mathcal{P} = \{\text{torso, L-arm, R-arm, L-leg, R-leg}\}$, giving a global frame feature z_g and part features $z_p \in \mathbb{R}^{P \times d}$. A 1-D conv VAE [25] with stride 4 emits $\mu, \log \sigma^2 \in \mathbb{R}^{L \times 256}$, $L = \lceil T/4 \rceil$. Stage-1 training is

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{bone}} \mathcal{L}_{\text{bone}}. \quad (1)$$

After 35k AdamW steps the tokenizer is frozen.

Fine-grained language–motion alignment. Global alignment remains symmetric InfoNCE on pooled CLIP and pooled latents, $\mathcal{L}_{\text{global}} = \text{InfoNCE}(\bar{c}, \bar{z})$, weight 0.25. On top of that, P learnable anatomical queries q_p attend to CLIP tokens,

$$\tilde{c}_p = \text{Attn}(q_p, c_{1:N}), \quad (2)$$

and are trained to match the corresponding part latent

$$\mathcal{L}_{\text{part}} = -\frac{1}{P} \sum_{p=1}^P \log \frac{\exp(\text{sim}(z_p, \tilde{c}_p)/\tau)}{\sum_j \exp(\text{sim}(z_p, \tilde{c}_j)/\tau)}. \quad (3)$$

The same $\tilde{c}_p$ are concatenated to CLIP tokens as extra DiT context, so the alignment is used at sampling time, not only as an unused projection head. Intuitively, “raises his left arm then kicks with his right leg” should bind to the left-arm and right-leg latents rather than to one pooled code.

Kinematic rectified flow. Given clean μ (detached), $t \sim \mathcal{U}(0, 1)$, $\varepsilon \sim \mathcal{N}(0, I)$,

$$z_t = (1 - t)\varepsilon + t\mu, \quad v^* = \mu - \varepsilon. \quad (4)$$

An 8-block DiT (width 512, 8 heads) predicts $v_\theta(z_t, t, c)$ with AdaLN modulation on the pooled text vector, self-attention on latent tokens, and cross-attention to CLIP+part tokens. Condition dropout 0.1 enables CFG [12]. The flow loss is $\mathcal{L}_{\text{RF}} = \|v_\theta - v^*\|_2^2$. We additionally decode the endpoint estimate $\hat{z}_1 = z_t + (1 - t)v_\theta$ with the frozen decoder and penalize motion-space kinematics:

$$\mathcal{L}_{\text{vel}} = \frac{1}{T-1} \sum_t \|(\hat{x}_{t+1} - \hat{x}_t) - (x_{t+1} - x_t)\|_1, \quad (5)$$

$$\mathcal{L}_{\text{bone}} = \sum_{(i,j) \in E} \left| \|\hat{p}_i - \hat{p}_j\|_2 - \|p_i - p_j\|_2 \right|, \quad (6)$$

plus an ℓ_1 term $\mathcal{L}_{\text{foot}}$ on the four foot-contact channels. The generator objective is

$$\mathcal{L} = \mathcal{L}_{\text{RF}} + \lambda_g \mathcal{L}_{\text{global}} + \lambda_p \mathcal{L}_{\text{part}} + \lambda_v \mathcal{L}_{\text{vel}} + \lambda_b \mathcal{L}_{\text{bone}} + \lambda_f \mathcal{L}_{\text{foot}}. \quad (7)$$

Sampling is Euler, $z \leftarrow z + \frac{1}{N}(v_\theta + \omega(v_c - v_\theta))$, with $N \in \{10, 20, 50\}$ and $\omega \in \{2.0, 2.5\}$.

Training. Stage 1: VAE only, 35k steps, effective batch 16, bf16. Stage 2: freeze graph+VAE, train flow+alignment, 100k steps, lr 2×10^{-4} , EMA 0.9999. The public HumanML3D checkpoint evaluated in Table 1 is the graph tokenizer + global InfoNCE + rectified-flow DiT (best .pt, step 35k under a 10-step val). Part alignment and kinematic decoding are enabled in the GALA-v2 configuration; we do not back-fill Table 1 with unrun numbers.

4 Experiments

Protocol. Official HumanML3D test [1], 4544 clips, Guo / MDM evaluator [2], batch 32, 20 replications, mean $\pm$ 95% CI. Diversity is compared to real motion (9.503), not maximized. KIT-ML uses the same evaluator on the 251-D official split. Hardware: RTX 4060 8 GB. GALA has 59.9M parameters (DiT 53.6M, VAE 3.2M, graph 1.6M).

Literature rows in Table 1 for T2M / MDM / M2DM / EMDM / GENMO follow GENMO Table 4 [13]; MLD / T2M-GPT / MoMask follow MoMask [8]. GENMO is listed as a generalist reference, not as the comparison target.

4.1 HumanML3D

Fig. 2 and Table 1 share one reading.

Semantic metrics at low NFE. GALA-20 reaches R@3 0.768 and MM Dist 3.248 with 20 function evaluations—above MDM (0.611 / 5.566) and M2DM (0.763), and within 0.02 R@3 of MLD and T2M-GPT, which are heavier discrete or long-chain latent systems. Diversity 9.53 matches Real 9.50. R@1 / R@2 are $0.447 \pm .003$ / $0.654 \pm .003$ (GALA-20) and $0.428 \pm .003$ / $0.632 \pm .003$ (GALA-50).

FID is the remaining gap, and it is a generator gap. GALA-50 FID 0.312 sits between M2DM (0.352) and T2M-GPT / EMDM / MoMask (0.141 / 0.112 / 0.045). We do not claim SOTA fidelity. Val reconstruction FID is 0.003 (Fig. 3): the Guo extractor barely distinguishes reconstructed clips from real ones, so 0.31 generated FID is a flow-matching problem. Kinematic endpoint losses in Sec. 3 target that gap; they are not yet reflected in Table 1.

GENMO’s SMPL $\leftrightarrow$ HumanML3D conversion makes its FID 0.216 a weak target for a native-HML3D model. We report it for completeness and do not organize the paper around beating it.

4.2 KIT-ML

Table 2 uses the same Guo protocol on official KIT-ML. Literature numbers are from MoMask [8] and MDM [2]. GALA is trained with the 21-joint graph and the same v2 recipe (configs/gala_kitml_flow.v2.yaml).

4.3 What each module contributes

Reviewers need to know whether the graph and the alignment do anything. Table 3 is the component ablation: Base RF is a conv tokenizer with no InfoNCE; subsequent rows add the

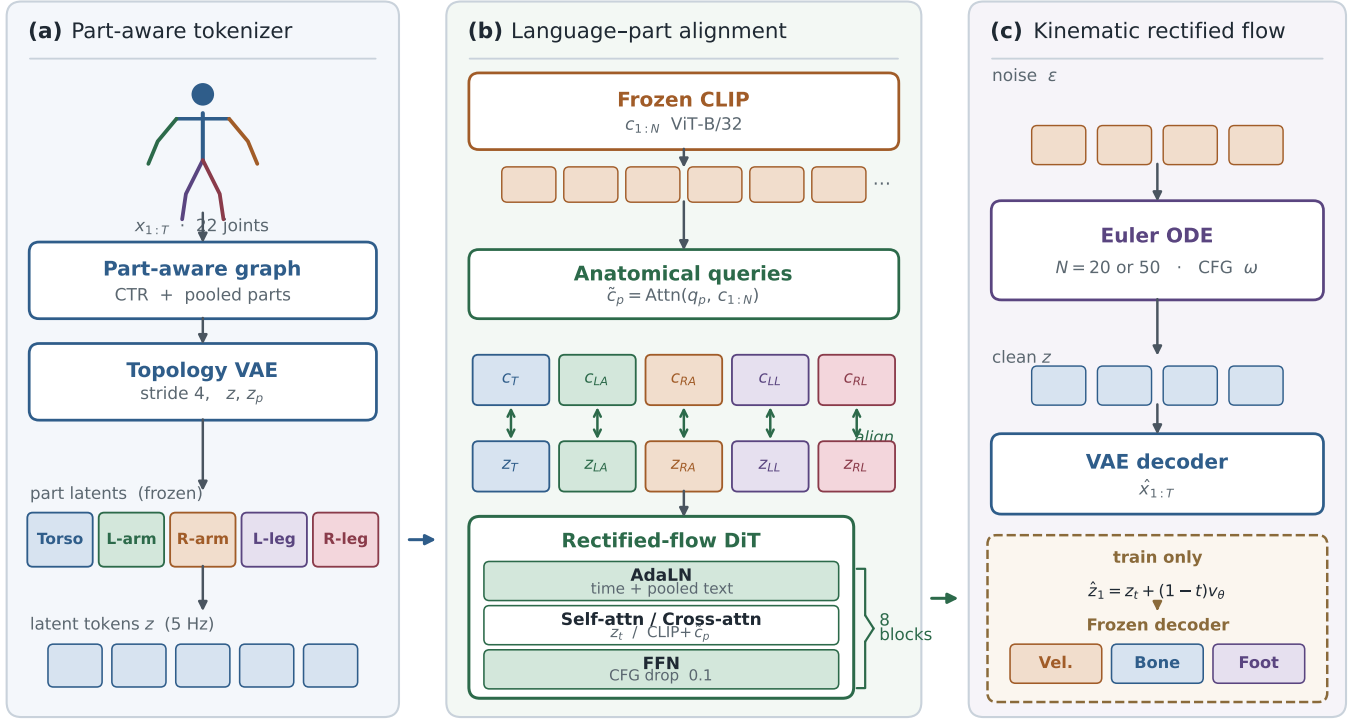


Figure 1: GALA, three causally linked stages. **(a)** A skeleton-graph VAE emits a 5 Hz latent and five part codes (torso, L-arm, R-arm, L-leg, R-leg). **(b)** Learnable anatomical queries attend to CLIP tokens and align to the matching part latents; $\tilde{c}_p$ condition the DiT. **(c)** Inference is a 20- or 50-step Euler ODE. A training-only branch decodes $\hat{z}_1$ and penalizes velocity, bone length, and foot contact. Code: <https://github.com/yanghxx/gala-motion>.

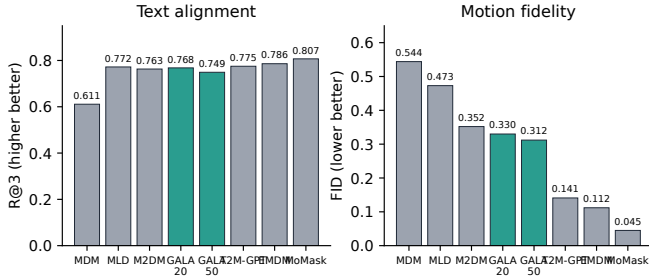


Figure 2: R@3 and FID on HumanML3D. Teal is GALA. Discrete / few-step SOTA (MoMask, EMDM) still lead FID; GALA’s advantage is 20-NFE semantic accuracy.

graph, global alignment, part alignment, and kinematic flow. The filled row is the public checkpoint (GALA-20). Empty cells are the same Guo protocol after the matching config finishes; we will not invent deltas.

4.4 Tokenizer: graph vs. conv

A reconstruction FID of 0.003 is not, by itself, evidence that the graph is necessary. Table 4 compares a graph-free conv VAE, an ST-GCN VAE (static adjacency), and the CTR VAE under the same stride and optimizer budget. The conv control is worse on FID (0.012 vs. 0.003) and MPJPE (0.105 vs. 0.080); a static ST-GCN already matches CTR reconstruction FID, while CTR further cuts MPJPE (0.088→0.080). The graph is

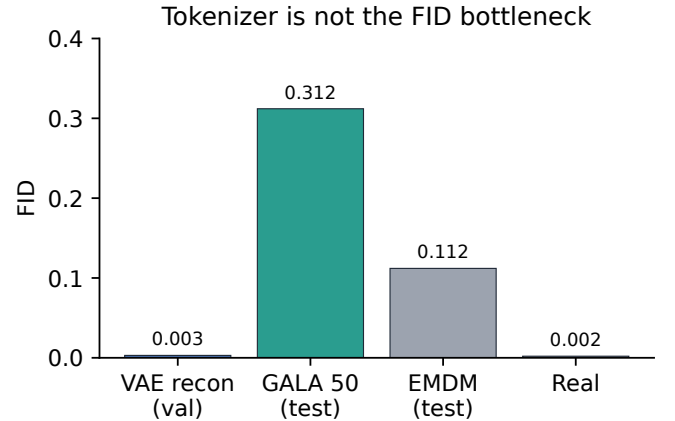


Figure 3: Val VAE reconstruction FID versus generated test FID. The tokenizer floor is far below the generator; optimizing the VAE further is the wrong spend.

therefore not a no-op; the adaptive topology is a smaller but measurable joint-error gain.

4.5 Sampling and the quality–efficiency Pareto

A 3×4 val scan on the frozen public checkpoint (Table 5) shows that $\text{CFG} = 1.0$ collapses both FID and R@3, while 20 steps / CFG 2.5 maximizes R@3 and 50 steps / CFG 2.0 minimizes FID (val 0.290), transferring to test as 0.312 / 0.749. On one RTX 4060, GALA-20 produces a $T=196$ clip in 140 ms (7.1

Table 1: HumanML3D test, Guo protocol. GALA is 20 replications on 4544 clips. $\uparrow$ higher, $\downarrow$ lower, $\rightarrow$ closer to Real. GALA-20/50: graph tokenizer + global alignment + rectified flow.

Method	Rep.	NFE $\downarrow$	R@3 $\uparrow$	FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
Real [1]	HML3D	–	0.797	0.002	2.974	9.503
T2M [1]	HML3D	–	0.740	1.067	3.340	9.188
MDM [2]	HML3D	1000	0.611	0.544	5.566	9.559
MLD [6]	HML3D	50	0.772	0.473	3.196	9.724
M2DM [9]	HML3D	–	0.763	0.352	3.134	9.926
T2M-GPT [7]	HML3D	–	0.775	0.141	3.121	9.761
EMDM [17]	HML3D	10	0.786	0.112	3.110	9.551
MoMask [8]	HML3D	–	0.807	0.045	2.958	–
GENMO [13]	SMPL	–	0.632	0.216	3.466	11.342
GALA, 20 steps, CFG 2.5	HML3D	20	0.768 $\pm$.002	0.330 $\pm$.011	3.248 $\pm$.008	9.529 $\pm$.085
GALA, 50 steps, CFG 2.0	HML3D	50	0.749 $\pm$.003	0.312 $\pm$.009	3.342 $\pm$.009	9.365 $\pm$.115

Table 2: KIT-ML, Guo protocol. GALA row is filled after the official 20-replication run.

Method	R@3 $\uparrow$	FID $\downarrow$	MM Dist $\downarrow$	Diversity $\rightarrow$
Real	0.779	0.031	2.788	11.08
T2M [1]	0.681	3.022	3.488	–
MDM [2]	0.396	0.497	9.191	10.85
MLD [6]	0.734	0.404	3.204	–
T2M-GPT [7]	0.745	0.514	3.007	–
MoMask [8]	0.781	0.204	2.779	–
GALA	–	–	–	–

Table 3: Component ablation on HumanML3D (target protocol: GALA-20, 20 replications). The filled row is the current official checkpoint.

Model	Graph	Glob.	Part	Kin.	R@3 $\uparrow$	FID $\downarrow$
Base RF					–	–
+ Graph	✓				–	–
+ Global	✓	✓			0.768	0.330
+ Part	✓	✓	✓		–	–
GALA-v2	✓	✓	✓	✓	–	–

clips/s) and GALA-50 in 340 ms; a 9.8 s motion is therefore generated far faster than real time. Table 6 times only GALA on this GPU; MDM / EMDM latency stays blank until those codebases are re-run at batch 1, $T=196$, because NFE is not wall-clock time.

5 Conclusion

GALA is a topology-aware continuous generator whose parts are independently testable: a nearly lossless graph tokenizer, language–part alignment beyond pooled InfoNCE, and a kinematic rectified flow that is cheap to sample. On official HumanML3D, 20 Euler steps give R@3 0.768 and real-like diversity on an 8 GB GPU. FID 0.31 is not competitive with EMDM, T2M-GPT, or MoMask; the tokenizer is not the bottleneck, so the remaining spend belongs on the generator (Table 3) and on KIT-ML under the same Guo protocol. We did not re-run

Table 4: Tokenizer reconstruction on HumanML3D val (1504 clips, no sampling).

Tokenizer	Recon FID $\downarrow$	MPJPE $\downarrow$	Bone $\downarrow$	Vel. $\downarrow$
Conv-VAE	0.012	0.105	0.061	0.039
ST-GCN-VAE	0.003	0.088	0.050	0.038
CTR-Graph-VAE	0.003	0.080	0.051	0.038

Table 5: Val sampling scan, 1504 clips, one seed. Official test uses the starred row.

Steps	CFG	FID $\downarrow$	R@3 $\uparrow$	MM Dist $\downarrow$
10	1.0	1.233	0.628	4.062
10	2.5	0.382	0.771	3.204
20	2.5	0.335	0.781	3.164
50	2.0	0.290	0.753	3.301
50	2.5	0.322	0.767	3.177

baselines; literature rows are copied from the cited tables.

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Table 6: Quality–efficiency operating points. GALA is 59.9M parameters. Latency: RTX 4060, batch 1, $T=196$, 10-run mean.

Method	NFE ↓	Params	Latency	R@3 ↑	FID ↓
MDM [2]	1000	–	–	0.611	0.544
MLD [6]	50	–	–	0.772	0.473
EMDM [17]	10	–	–	0.786	0.112
GALA-50	50	59.9M	340 ms	0.749	0.312
GALA-20	20	59.9M	140 ms	0.768	0.330

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